

# Ryne Starnes

Tallahassee, FL · 214-783-9511 · rynestarnesbusiness@gmail.com

## EDUCATION

Florida State University

Tallahassee, FL

Bachelor of Science in **Physics & Astrophysics**

August 2023 – May 2027

Bachelor of Science in **Applied & Computational Mathematics**

## WORK EXPERIENCE

Undergraduate Research Assistant - *Florida State University*

February 2024-Present

- Designed a neural network through Python that analyzes large datasets to compute a ‘di-Higgs cross-section’ given certain parameters under Dr. Harrison Prosper
- Modified a ‘Physics-informed neural network’ to solve complex ODEs more efficiently
- Used pandas library to refine raw Higgs data into normalized and organized data

ASSURE Student Intern – *University of Wisconsin-Madison*

June 2026-August 2026

- Working on automated tools for quality assurance in MRI machines
- Using machine learning to find hidden predictors of MRI degradation

Learning Assistant – *Florida State University*

January 2025-May 2025

- Facilitate student understanding of fundamental physics concepts, including 2D kinematics, forces, and energies
- Helped lab set-up, held regular office hours to assist with homework solutions and clarify difficult concepts

## INVOLVEMENT

FSU Society of Physics Students - *Marketing Director, Secretary*

August 2023-Present

- Managed various social media accounts for the club; designed graphics to advertise meetings and posters to advertise the club around campus.
- Led logistics and coordination for the chapter’s attendance to the 2025 Physics & Astronomy Congress and several STEM outreach events for the Tallahassee community.

Society of Physics Students – *Associate Zone Councilor*

June 2026-Present

- Connecting SPS chapter presidents of Zone 6 for greater collaboration and increased outreach efforts
- Organizing a regional research conference

## CONFERENCE PRESENTATIONS

- “Predicting a di-Higgs Mass through the Use of Neural Networks”, Florida Undergraduate Research Conference, Feb. 2025; Physics & Astronomy Congress, Oct. 2025; SPS Zone 6 Conference, Mar. 2026

## SKILLS

- Programming: Python, C++, UNIX
- Machine Learning: PyTorch, Sci-kit